

Digital Systems Pre-Report 1

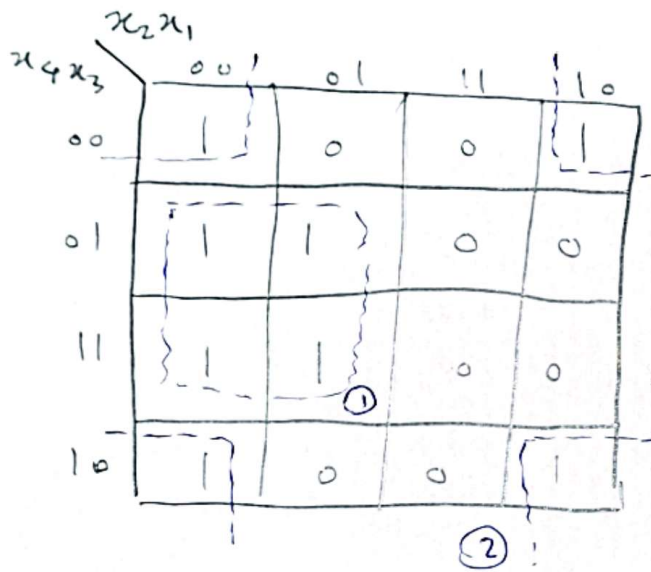
Arshiya Noorpoor

403102871

①

$$f = \sum (0, 2, 4, 5, 8, 10, 12, 13)$$

Karnaugh map



x_4	x_3	x_2	x_1	m
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

① $\rightarrow x_1, x_4 \rightarrow \overline{x_2} x_3$

② $\rightarrow \overline{x_4}, x_2 \rightarrow \overline{x_1} \overline{x_3}$

$$\rightarrow f = \overline{x_2} x_3 + \overline{x_1} \overline{x_3} = \overline{\overline{\overline{\overline{x_2} x_3} \cdot \overline{\overline{\overline{x_1} \overline{x_3}}}}}$$

